**Mid Semester Examination**

Winter 2026, Network Science

21st February 2026; Time: 50 Minutes [**40 Points Maximum**]

**SET A**

**1. Choose the correct alternative(s).** ***Negative points for wrong answers.*** **The answer is deemed wrong even if one answer is incorrect.** $[20 \times +1/-1 = +20/-20]$

(1) What term best describes a system that consists of a large number of sophisticated functional elements intricately connected to perform specific tasks that subsets of the system cannot execute?

(a) Modular System
(b) Decentralized System +1
(c) Complex System
(d) Isolated System

(2) Which are the two forces that helped the *emergence of network science*?

(a) Sparseness & Complexity
(b) Complexity & Ubiquitous nature of networks +1
(c) Availability of network maps & Universality of network characteristics
(d) Completeness of network maps & Availability of large networks

(3) How many units of memory would a network with '$n$' nodes need when stored as an 'adjacency matrix'?

(a) $n$
(b) $n^2$ +1
(c) $n^2 - n$
(d) $n^2 + n$

(4) The average degree ($<k>$) for a simple, fully connected graph with $n$ nodes would be

(a) $n$
(b) $n - 1$ +1
(c) $n^2$
(d) $\sqrt{n}$

(5) The time complexity of the Breadth First Search (BFS) algorithm is

(a) $O(N + L)$
(b) $O(L)$ +1
(c) $O(N)$
(d) $O(N^2 + L)$

(6) Which of the following is ***not*** an undirected graph?

(a) World Wide Web
(b) Actor Network +1
(c) Co-authorship Networks
(d) Protein interaction networks

(7) Which of the following is a characteristic of a sparse network?

(a) $L \gg L_{max}$
(b) $L = L_{max}$ +1
(c) $L \ll L_{max}$
(d) $L = 0$

(8) In a directed graph, a *sink* is a node with—

(a) $k_{in} = k_{out}$
(b) $k_{in} = 0$ +1
(c) $k_{in} \leq k_{out}$
(d) *None of the above*

(9) For a 'complete graph' (undirected), the number of edges in the network ($L$) is—

(a) $N - 1$
(b) $N$
(c) $N^2$
(d) $N(N-1)/2$ +1

(10) For which of the following conditions is a Random Network said to have achieved the '*critical point*'?

(a) $< k > \gg 1$

(b) $< k > = 0$ ✓ (+1)

(c) $< k > = 1$

(d) $< k > = N$

(11) In the error and attack tolerance experiments on random and scale-free networks study, the following observation was made regarding the 'Size of the Giant Component ($S$)':

(a) The $S$ for random graphs breaks down quicker with random errors than with targeted deletions.

(b) The $S$ for scale-free graphs is robust for targeted deletions as compared to random errors.

(c) The $S$ for scale-free graphs is intact for as many as 80% nodes deleted randomly. ✓ (+1)

(d) The $S$ for random graphs is intact for as many as 50% of nodes deleted randomly.

(12) In the Indian Railways Network research article, how is an edge between two stations defined?

(a) Two stations are said to be connected if a physical railway line connects them.

(b) Two stations are said to be connected if they are geographically proximate. ✓ (+1)

(c) Two stations are said to be connected if a train passes through them.

(d) Two stations are said to be connected if a train passes through them and stops at both stations.

(13) Which of the following justifies the correct statement regarding the links of a 'directed network'?

(a) At least one link is undirected

(b) At least two links are directed ✓ (+1)

(c) All links are directed

(d) All but one link is directed

(14) In an simple, undirected network, if the shortest path between node $i$ and node $j$, $L_{ij} = 3$ is , then which of the following must be true?

(a) There are exactly three distinct paths between $i$ and $j$.

(b) The shortest path between $i$ and $j$ uses exactly three links. ✓ (+1)

(c) There is no path shorter than three links between $i$ and $j$.

(d) Both (b) and (c).

(15) Two nodes in a directed network $L_{ij} = 2$ but $L_{ji} = \infty$. What does this imply?

(a) A path exists from $j$ to $i$, but no path exists from $i$ to $j$.

(b) The network is undirected. ✓ (+1)

(c) A path exists from $i$ to $j$, but no path exists from $j$ to $i$.

(d) The network has a bridge.

(16) As the probability $p$ of two nodes being connected **increases** in a random network model—

(a) the network becomes more disconnected.

(b) the network tends become more connected. ✓ (+1)

(c) the degree distribution becomes uniform.

(d) the weight of the edges in the network increases.

(17) In the 'evolution of a random network', which regime has mostly isolated nodes and tiny clusters?

(a) Connected regime

(b) Supercritical regime

(c) Subcritical regime (+1)

(d) Clustered regime

(18) The shortest path (geodesic) in a network refers to:
(a) The path with the most nodes
(b) The path containing only weighted edges
(c) The path minimizing the number of edges (or summed weight) between two nodes (+1)
(d) A path that spans the network's diameter

(19) A 'bipartite network' is best described as:
(a) A network with two types of nodes, where edges only connect across types
(b) Any network with an even number of nodes
(c) A network where every node connects to every other node (+1)
(d) A network with hierarchical layers

(20) Which condition must a network satisfy to allow a Eulerian path (traverse each edge exactly once)?
(a) All nodes have even degree
(b) Exactly zero or two nodes have odd degree (−1)
(c) Every node has at least one edge
(d) The network must be fully connected

2. For the following real-world examples of networks, clearly state what is a node, an edge and what constitutes the weight/direction, if any. [3×1=3]

**(a) Bipartite Network — Diseasome:**

Nodes (Type 1): Genes (1) Nodes (Type 2): Diseases

Edges: Indicates which genes are responsible for which diseases. Weights/Direction: Undirected, unweighted

**(b) Weighted Networks — Facebook**

Nodes: Users or people (1) Edges: Facebook connection or friend

Weights/Direction: undirected, ~~unweighted~~ Weighted: for denoting level of acquaintance/ friendship and interactions.

**(c) Directed Network — Airport Networks**

Nodes: Airports (1) Edges: Indicates whether 2 airports are connected via a direct flight according to the weekly timetable of domestic airplanes.

Weight/Direction: Directed (can be made ~~do~~ undirected since most are bidirectional), weights are no. of flights b/w airports in a week.

3. In the example of networks related to Food and Recipes discussed in the 'Network Science' book, describe what are the (a) mono-partite, (b) bi-partite and (c) tri-partite networks that one could construct. Clearly explain each of these networks along with their node and edges. [3×1=3]

Compounds.

4. When implementing the Watts and Strogatz strategy, what is the condition/constraint on the 'number of nodes ($n$)' for generating a $k$-regular graph? [1]

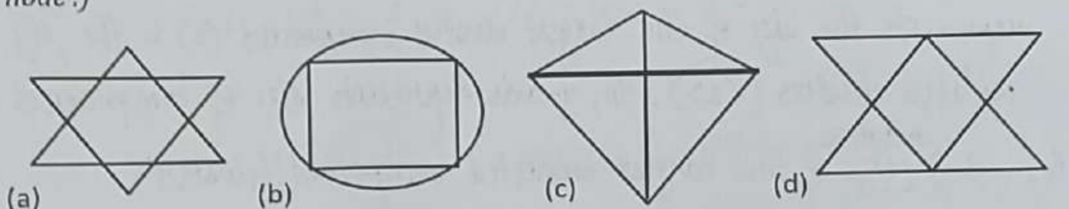
We need k (degree) to be lower than n (no. of nodes). So, k<n.
Also, by the hand shaking lemma, we know sum of degrees should be equal to 2* no. of edges.
So, n × k = 2 × L, so, n×k should be even, where L ⇒ no. of edges.

1

5. In a scale-free network with degree exponent $\gamma = 3$ (i.e. $P(k) = k^{-3}$), prove that nodes with degree $k = z$ are approximately eight times more populated than proteins with $k = 2z$. [1]

So, given P(k) = k-3

6. State and prove the Euler's Theorem in connection with Königsberg's bridge-crossing problem. In which of the following graph(s) one could trace an Euler circuit so as to start from a given node and return to the same node by tracing every edge exactly once; without having to retrace any of the edges? List your answers with graph-theoretic justification. *(Please note that every intersection of lines denotes a 'node'.)* [1+2=3]



(a) (b) (c) (d)

~~We can prove by contradiction.~~

a) Degree distrib : nodes with $k = 2 \Rightarrow 6$, nodes with $k = 4 \Rightarrow 6$. 1
All are even degree nodes. So, euler circuit is possible.

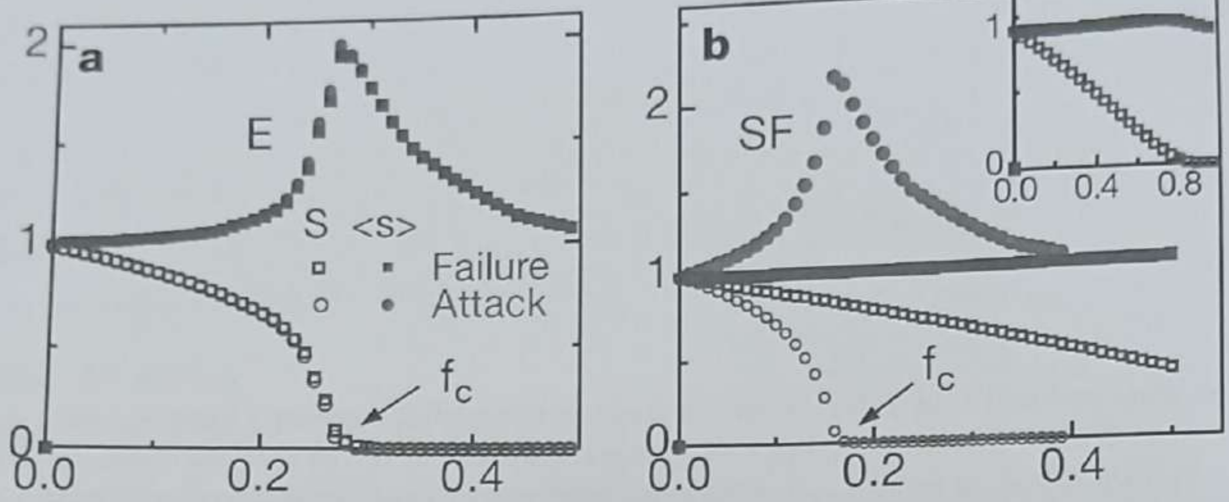
b) Nodes with degree $k = 4 \Rightarrow 4$. All are even degree nodes. So, euler circuit is possible.

c) Nodes with degree $k = 4 \Rightarrow 1$, $k = 3 \Rightarrow 4$, more than 2 odd degree nodes, no euler circuit possible.

2 d) Nodes with $k = 2 \Rightarrow 4$, $k = 4 \Rightarrow 4$, all are even degree nodes, so euler circuit possible.

Proof: Since odd degree nodes have to be at the starting & ending points of the eulerian circuit, the no. of odd degree nodes cannot be more than 2, **

7. Briefly explain results, depicted in the following figures, obtained by Albert *et al.* in their studies on 'error and attack tolerance of complex networks'. (a) What is represented on the x- and y-axis? (b) What do S and <s> stand for? (c) What are the conclusions from each data curve in figure **a** and figure **b**? (d) What is $f_c$? (e) What does the *inset* (smaller plot on the right-top) in figure **b** imply? [**4**]


a
b
E
SF
S
<s>
Failure
Attack
$f_c$
0
1
2
0.0
0.2
0.4
0.8

8. Let $A$ be the $N \times N$ adjacency matrix of an undirected unweighted network, without self-loops. Let 1 be a column vector of $N$ elements, all equal to 1. In other words $1 = (1, 1, ..., 1)^T$, where the superscript $T$ indicates the *transpose* operation. Use the matrix formalism (multiplicative constants, multiplication row by column, matrix operations like transpose and trace, etc., but avoid the sum symbol $\Sigma$) to write expressions for:

**[5×1=5]**

(a) The vector k whose elements are the degrees $k_i$ of all nodes $i = 1, 2, ..., N$.

(b) The total number of links, $L$, in the network.

(c) The number of triangles $T$ present in the network, where a triangle means three nodes, each connected by links to the other two (Hint: you can use the trace of a matrix).

(d) The vector $k_{nn}$ whose element $i$ is the sum of the degrees of node $i$'s neighbors.

(e) The vector $k_{nnn}$ whose element $i$ is the sum of the degrees of node $i$'s second neighbors.